



CSE 360: Computer Interfacing
Lab Report (Section 06)

Title: **Semi-Outdoor Climate Control System Using Automated Cooling And Humidification.**

Prepared By:

Sabrina Rahman Mazumder -20301218

Nabila Samroj Oishe- 20201216

Md. Shoaibur Rahman-21101322

Md. Abu Hussain -21101242

Mithila farzana Mila-22101709

Submitted Date: 03-05-2025

1. Abstract:

Most people spend a large part of their day outdoors, so creating a comfortable environment for an open place has gained prime importance. It's also more concerned with creating a comfortable environment for a large indoor or outdoor open space where customary cooling systems are expensive. The temperature and humidity levels can take an abrupt turn because of different weather conditions and manual measures like turning on fans or humidifiers prove to be tedious and ineffectual. This research aims to solve that problem by developing an automated, intelligent cooling system that responds to the current ambient conditions with minimizing pricing cost. The apparatus uses DHT11 sensors to measure temperature and humidity, and with specific thresholds being reached, Arduino Uno serves as the major controller to turn on a fan or water pump. An I2C-based LCD panel expressly shows all current readings, providing real-time feedback. By only running when essential the design not only effectively maintains interior comfort but also cuts down on wasteful energy consumption. This project serves as both an actual experiment and an instructional tool, showcasing the collaboration of sensors, microcontrollers, and basic automation in an interactive system with embedded components.

2. Introduction

We proposed here to design an Automated Cooling and Humidification System that will scan and respond to environmental surroundings, thereby enhancing indoor or outdoor

open space easement by conditions in real-time. The system will use Arduino Uno as the base controller and estimate temperature and humidity readings through one or more DHT11 sensors and automatically operate a fan or mist pump when programmed threshold limit conditions are exceeded. Current measurements will be shown on a 16x2 I²C LCD for a translucent outline of the state of the system. Present measurements will be displayed on a 16x2 I²C LCD for clear visibility of the status of the system. High-current devices can be safely managed by Arduino, relays guaranteeing safety, I²C and digital protocols making sure efficient communication among components. The system not only improves energy efficiency by operating when needed , but also makes it a real application for embedded systems, sensor interfacing, and basic automation, thus making it a standard for using microcontrollers.

2.1 Problem Statement

As human beings, we spend the majority of our time indoors so it's necessary to maintain a pleasant indoor/ outdoor climate. Indoor comfort and air quality can easily be affected by outdoor temperature and humidity fluctuations. So turning on fans or pumps manually is a waste of time. A system that senses both indoor and outdoor environmental conditions and gives power to a fan and misting pump that can dynamically regulate comfort and also reduce energy loss. The project proposed will show an integrated climate-control solution.

2.2 Objectives:

- Track Environmental Conditions: DHT11 sensors will read temperature and humidity levels indoors (optionally outdoors) in real time.
- Automatic Control: Switch on the pump/fan automatically if readings exceed set parameters (fan if temperature crosses the set and pump if humidity below the set).
- Display Data: Current humidity and temperature shown on an I²C LCD.

- **Robust Interfacing:** Offer robust communication between sensors, actuators, display, and the Arduino Uno R3.
- **Configurable Thresholds:** Enable simple modification of temperature/humidity thresholds in software for varying conditions or demos.

2.3 Significance:

- **Comfort and Health:** An automatic climate control system monitors temperature and humidity which helps preserve comfort and can be a solution to problems such as overheating or excessive dryness.
- **Energy Efficiency:** Using ventilation, heating, and air conditioning uses a significant amount of electricity. Smart automation can save huge amounts of energy by operating equipment only when required.
- **Real-time Awareness:** The LCD delivers real-time environmental and system status information feedback, offering enhanced user awareness.

3. Interfacing Design

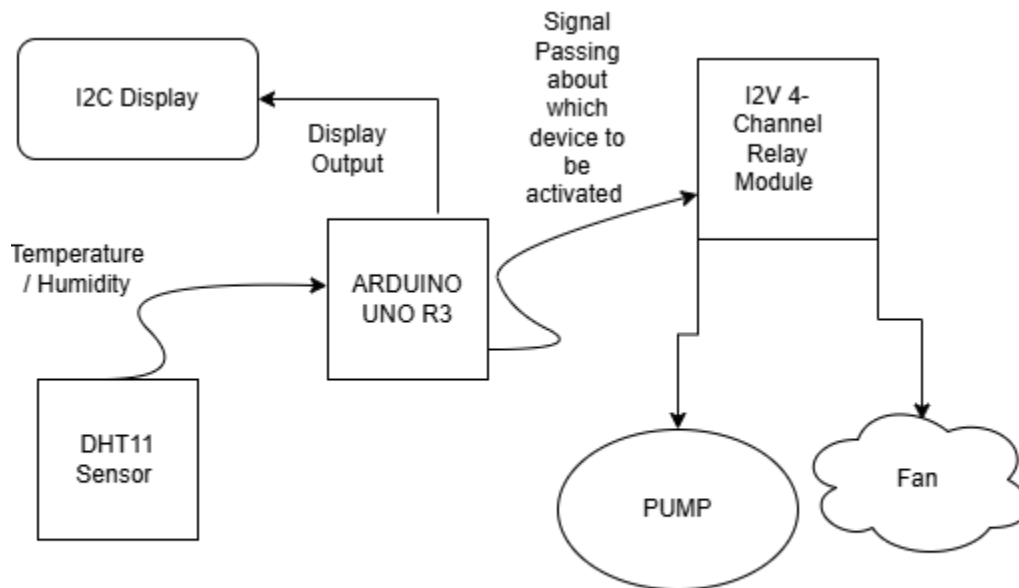


Figure: Interface Design

3.1 Interfacing Components:

Electric Components :

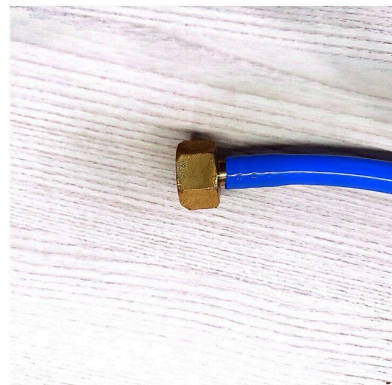
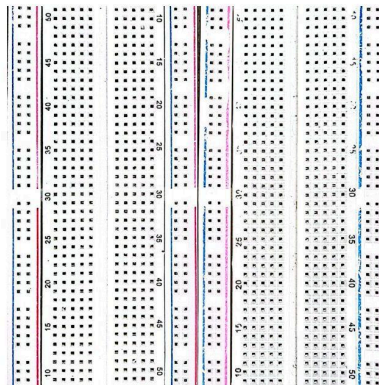
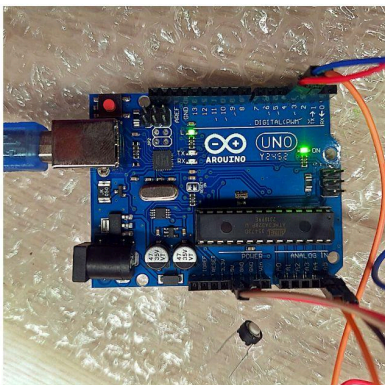
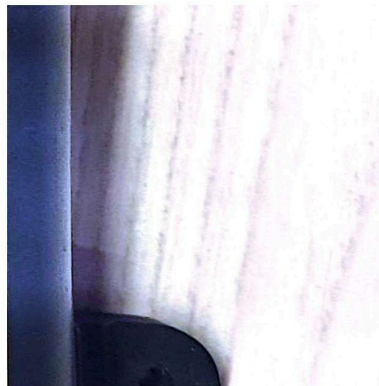
- **Arduino Uno R3:** It actually serves as the central controller, reading sensor data and driving actuators.
- **DHT11 Sensor:** It provides calibrated readings via a digital interface single wired for Digital temperature and humidity sensor;.
- **I2C LCD Display:** 16×2 character display module; communicates over I2C lines (SDA, SCL) to minimize Arduino pin usage.
- **12 V 4-Channel Relay Module (Opto-Isolated):** Switches high-voltage loads (pump, fan) using low-voltage Arduino signals; optical isolation protects the microcontroller from voltage spikes.
- **Pump:** Delivers water for forming mist; powered through one relay channel.
- **Fan:** Provides airflow for cooling or ventilation; powered through a separate relay channel.

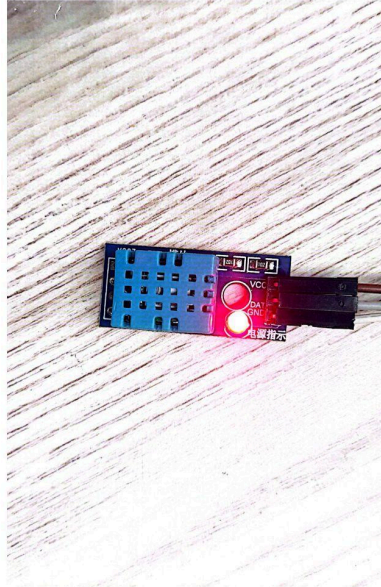
Hardware components :

- **Brass joint:** it is for motor and pipe joint
- **High pressure pipe :** it works for water out.
- **Nozzle :** It joints the nozzle and pipe for mist making .
- **Nozzle Joint:** It is actually connected to the nozzle and pipe.
- **Pump:** It will take the water and this water turns out mist.

- **Pump Adapter** : It is actually connection point for joining the Pipe and Pump

3.2 Images of Components





3.3 System Block Diagram



Figure : Full Setup of our project

3.3 Interfacing Challenges:

	Challenge	Solution
--	-----------	----------

Signal Integrity & Noise	Switching inductive loads (pump, fan) can introduce voltage spikes on shared grounds.	Using the relay's opto-isolation and add flyback diodes or snubber circuits across the relay outputs.
Power Supply Stability	The Arduino logic is 5 V, while the relay coils require 12 V so it may sag when driving multiple modules	Providing a dedicated 5 V regulator or using a 5 V switching power supply capable of supplying ≥ 1 A, and a separate 12 V supply for relays and actuators may help.
Component Compatibility	The DHT11's timing-sensitive protocol must be precisely handled by the Arduino.	Leverage an existing DHT library with interrupt-safe code and ensure proper pull-up resistor (~ 4.7 k Ω) on the data line.
I2C Bus Reliability	Long wiring runs to the display can degrade I2C signals.	Use twisted-pair cables for SDA/SCL, add pull-up resistors (4.7 k Ω), and keep bus length under 1 m where possible

Communication Protocols:

Protocol	Devices	Role
I2C	LCD Display	Two-wire bus (SDA, SCL) for display updates; conserves

		Arduino pins for other I/O, supports multiple devices on the same bus by address; the bus operates at standard speeds which is more than sufficient for updating the display.
Single-Wire Digital	DHT11 Sensor	One-wire custom timing protocol; simple data transfer of temperature/humidity ; uses precise timing.
GPIO Digital Outputs	Relay Module	Simple HIGH/LOW signals to power relay inputs; static logic levels, no clock protocol;

4.Protocol Justification:

I2C for Display:

- Only two lines are required for any number of I2C peripherals. The other digital pins for sensors and relays.

- It also has the ability to chain multiple I2C devices on the same bus.
- It data rates till 100 kbit/s in the standard mode then for the fastest mode 400 kbit/s. Next high speed mode for 3.4 Mbit/s.
- It takes data to be transferred when it is affected by the data rate and length transfer.
- Total bus length using it as a standard speed is 50 cm.

Single-Wire for DHT11:

- Mature Arduino libraries handle all the precise timing, error checking, and CRC validation.
- DHT11 only needs a single data wire (plus power/ground) to read. It takes very little wiring effort, a few lines of code and negligible CPU time.
- It has a range 0 to 50 degree Celsius and a lower temperature limit is reached when it reports 1 degree Celsius.
- It responds approximately in 1 second and the entire communication process takes around 4ms.

Digital GPIO for Relays:

- The Arduino can immediately set a pin HIGH to trigger the relay's opto-input, causing the relay to close and power the fan/pump. This direct approach has minimal latency and is fully deterministic.
- It also avoids the overhead of serial communication protocols, which are unnecessary for such basic control.
- It refers to when digital signals are sent to the relay and it is actually working like a control like a switch for pump and fan.

5.Implementation Plan

5.1 Methodology

The I2C LCD acts as a real-time display module to showcase temperature and humidity of the environment so it gives a clear idea of how the overall misting system has an impact on cooling

the environment and how these metrics are constantly monitored for operating the overall device. It uses LiquidCrystal_I2C.h library made by Adafruit.

```
#include <LiquidCrystal_I2C.h>
#define LCD_ADDR 0x27
#define LCD_COLS 16
#define LCD_ROWS 2
LiquidCrystal_I2C lcd(LCD_ADDR, LCD_COLS, LCD_ROWS);

lcd.setCursor(0,0);
  lcd.print("T:");
  lcd.print(temp, 1);
  lcd.write(byte(223)); // ° symbol
  lcd.print("C ");
  lcd.print("H:");
  lcd.print(hum, 0);
  lcd.print("%");
```

The relay device acts as an automated power switch which turns on or off a device, whether it is the fan or the pump based on a certain threshold of temperature and humidity of the environment. It uses basic logical if-else statement.

```
String activeState = "";
// RELAY1: cooling fan if temp > high threshold
if (temp >= TEMP_HIGH_THRESHOLD) {
  digitalWrite(RELAY1_PIN, LOW);
  activeState += "Fan Activated.";
} else {
  digitalWrite(RELAY1_PIN, HIGH);
}

// RELAY2: mist if humid < high threshold
if (hum <= HUM_HIGH_THRESHOLD) {
  digitalWrite(RELAY2_PIN, LOW);
  activeState += "Mist Activated."; // clear old text (pad to LCD_COLS)
} else {
  digitalWrite(RELAY2_PIN, HIGH);
}
```

```
    }  
    lcd.setCursor(0, 1);  
    lcd.print(activeState);  
}
```

The DHT11 gathers readings and information from the environment of its temperature and humidity using its built-in sensors. It uses a simple DHT.h library for this purpose.

```
#include <DHT.h>  
#define DHTPIN 2  
#define DHTTYPE DHT11  
DHT dht(DHTPIN, DHTTYPE);  
// Read DHT  
float hum = dht.readHumidity();  
float temp = dht.readTemperature(); // °C  
  
// Validate  
if (isnan(hum) || isnan(temp)) {  
    lcd.clear();  
    lcd.setCursor(0,0);  
    lcd.print(" DHT11 Error ");  
    Serial.println("DHT11 read failed");  
    delay(2000);  
    return;  
}
```

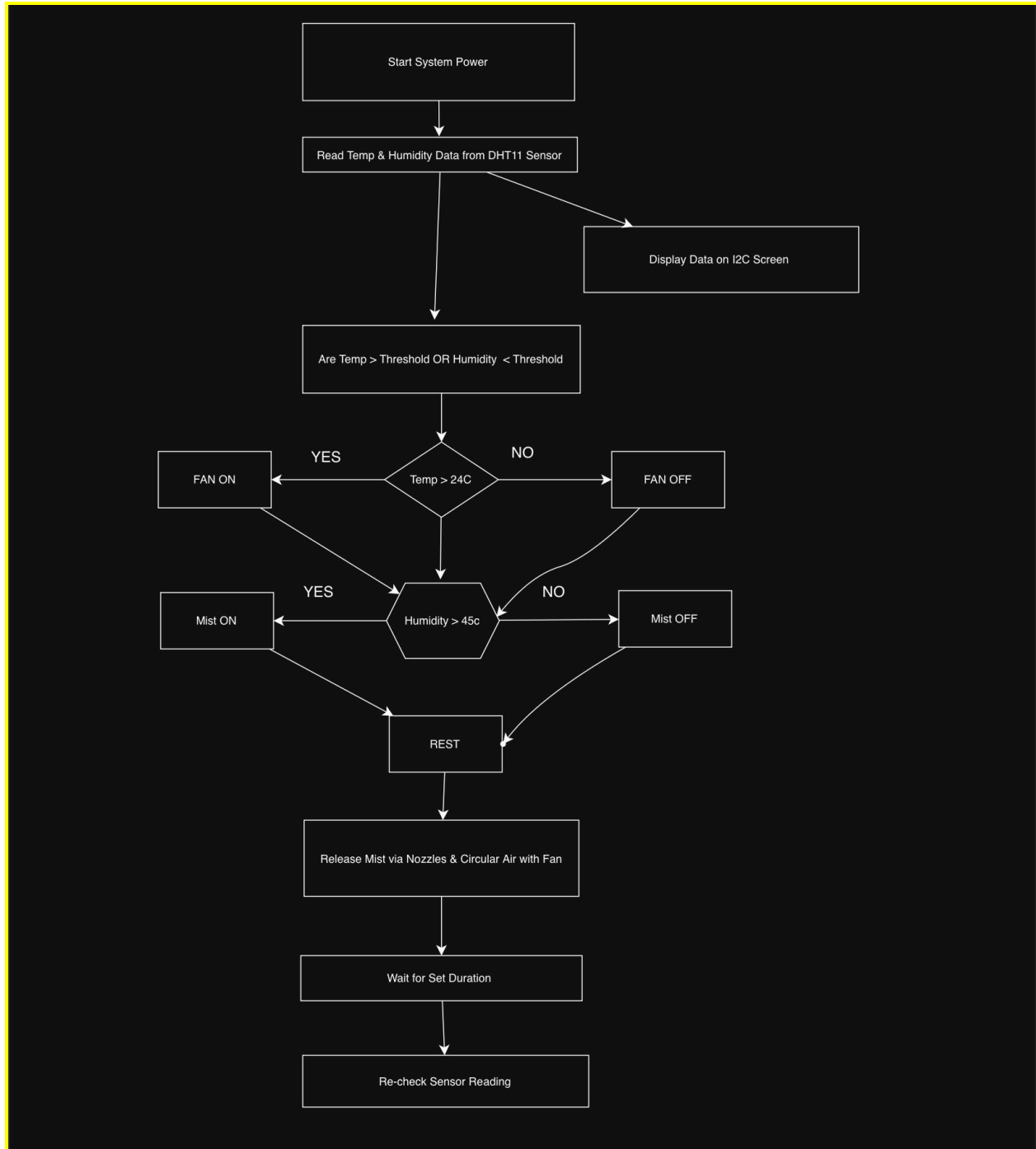


Figure: System interfacing flowchart

5.2 Expected outcome

i) It will cool the indoor and outdoor environment using mist. Depending on the humidity and temperature of the room, it will either turn on or off the components. By monitoring temperature and humidity, it will control the fan, and pump to cool the room while also preventing sweat.

ii) Aims to reduce issues related to heat, like heat strokes, high temperatures, resulting in discomfort for open places with large gatherings like BRAC University Cafeteria for example. With around 500 to 1000 people spending their leisure time in such places, we focus on their demographics to ensure their comfort.

iii) We're using sensors like DHT11 to monitor and gather temperature and humidity readings from the environment, where temperatures are read in Celsius and humidity in percentages. This is mainly used to understand the current ambient conditions before mist is applied. We've set a threshold of 24 degree celsius and 55% humidity. Mist starts when humidity reaches below 45% & fan starts when temperature reaches above 24 degree Celsius. This avoids unnecessary misting on cool humid days. It helps save water more efficiently. This also has a manual mode where the user can turn on or off the fan or mist at his will without the automation mechanism.

iii) Interfacing Design Reliability: This system is designed to function in both indoor and outdoor environments. First high pressure s mist nozzles (). Second, a fan to throw air flow rate 150 m³/min.

iv) This system is fully automated . It will work for turning on and off .

5.3 Obstacles :

Due to lack of equipment availability in Bangladesh, a lot of the equipment had to be outsourced from countries like China where delivery takes around 30 days, resulting in our inability to implement more advanced functionalities.

6. Future work and Potential Applications

6.1 Future work:

A simple yet efficient automatic climate control system has been showcased in the project model. Further upgrades would see its functionality, efficiency, and flexibility enhanced to a great level.

A solar power source can be introduced as a major upgrade. An extra solar panel with a charge controller would make this system work easily in any off-grid or power-deficient place when the main power source is absent. It would help good energy programs, and its portability would thus be increased for setup in distant places or even for outdoor use.

Another good upgrade is adding an air ionizer part. Ionizers make indoor open space air better by reducing airborne dirty things, allergens, and bacteria plus controlling how hot and humid it is. An ionizer would help give a more complete temperature and air quality management solution when added to the current fan and humidity controls. This is very important in open spaces of homes, hospitals, and factories where clean air is needed.

We can also implement a manual mode which will make all the components like fan and misting to be used manually by the flick of a switch without automation. This gives liberty for the user to still use the cooling peripherals at their own will.

6.2 Potential Applications:

This setup has many practical uses in homes, businesses, and schools. In modern houses, it might function as a piece of joined HVAC help machinery, automatically keeping cozy open space conditions and reducing power use. In places where plants are grown or inside farms, the system could assist in controlling moisture and heat levels to achieve the best growing environment; especially if it includes sunny-powered working. This endeavor is highly beneficial for teaching embedded systems, sensor interface, and real-time control in academic contexts. Due to its modular design, expansion or modification for better experimentation becomes fairly easy. This system can be applied as a cheap backup to keep an eye on and manage ambient conditions in commercial settings if the main system goes wrong, like in large open work areas or storage spaces. Next we have implemented in our university cafeteria's open space as in real problems at our university cafeteria where huge crowds stay into 9 to 5 p.m. During summertime when the temperature will increase then people get heat from their environment even if central fans work. To bring cool with cost effective products we introduced Integrated Outdoor weather Control system to raise air quality by Purifying air. Here we will say additional future practical suggestions.

References

1. Farnham, C., Yuan, J., & Emura, K. (2025). Evaluation of the Cooling Effect of an Outdoor Misting Fan for Workers in Hot Environments Wearing Personal Protective Equipment (PPE). *Clean Technologies*, 7(1), 9. <https://doi.org/10.3390/cleantechnol7010009>
2. Wang, P., Lu, S., Wu, X., Tian, J., & Li, N. (2024). Mist Spraying as an Outdoor Cooling Spot in Hot-Humid Areas: Effect of Ambient Environment and Impact on Short-Term Thermal Perception. *Buildings*, 14(2), 336. <https://doi.org/10.3390/buildings14020336>

